

Department of Geography & Environment
The George Washington University
Washington, DC 20052, USA

July 14, 2026

The Editor-in-Chief

Remote Sensing Applications: Society and Environment

Dear Editor-in-Chief,

On behalf of my co-authors, I am pleased to submit our manuscript, “Sparse Feature Selection, Not Deep Learning Capacity, Drives Spatial Transfer in Crop Classification,” for consideration as a research article in *Remote Sensing Applications: Society and Environment*.

Accurate crop maps underpin food-security monitoring and agricultural land-use planning across the Global South, yet operational deployment there almost always means applying a model to regions where no training labels were collected. This is where remote sensing must deliver societal value, and also where the field’s standard evaluation is weakest: most studies validate within a single region, measuring in-distribution fit rather than the spatial transfer that real monitoring demands. We confront this gap in a rain-fed, smallholder-relevant landscape of the Western Cape of South Africa, a regional case study whose implications generalize across data-scarce agricultural regions worldwide.

Using an openly available South African Sentinel-2 dataset (optical only, to maximize applicability where only optical data exist), we benchmark twenty-three classical and deep models both in-region and on a spatially disjoint holdout tile, and find that the two rankings *invert*: dense temporal deep networks that lead under conventional validation degrade sharply under transfer, while simpler, sparser models stay robust. Our contribution is mechanistic rather than merely empirical: we show that sparse, axis-aligned feature selection (not model capacity) governs whether a model survives spatial transfer, confirming it with a controlled capacity-versus-sparsity comparison and by grafting the bias onto a deep network (L-TAE-S) that achieves the best transfer performance at a fraction of the training cost.

We believe this work fits the aims of *Remote Sensing Applications: Society and Environment*: an application-driven, interdisciplinary study of agricultural land use and food-security monitoring that offers readers both a practical caution about how crop-classification models are evaluated before deployment and a concrete, transferable design principle for models that generalize to unlabeled terrain. The processed dataset is openly deposited on IEEE DataPort (DOI: 10.21227/n69k-r458) and all code is public. The manuscript is original, has not been published previously, and is not under consideration elsewhere; all authors approve the submission and declare no competing interests. Thank you for considering our work; we look forward to the reviewers’ feedback.

Sincerely,



Michael L. Mann
Department of Geography & Environment
The George Washington University
mmann1123@email.gwu.edu